□ Week 9 – DeFi, Tokenization & Stablecoins

□ Section 1: Introduction to DeFi (Decentralized Finance)

□ What is DeFi?

Decentralized Finance (DeFi) is a new financial system built on blockchain technology that removes traditional intermediaries like banks, brokers, and payment processors. Instead of trusting an institution, you trust **smart contracts**—self-executing programs on the blockchain that enforce rules automatically.

□ Key Characteristics of DeFi:

Feature	Description
Trustless	No need to trust banks; smart contracts handle funds automatically.
Permissionless	Anyone with an internet connection and a crypto wallet can access DeFi services.
Open Source	Most DeFi protocols publish their code publicly for anyone to audit.
Interoperable	DeFi applications can be composed together like "Money Legos."
Non-custodial	You control your funds at all times via your private keys.

□ DeFi vs CeFi (Centralized Finance)

Feature	DeFi	CeFi
Intermediaries	None	Banks, brokers, custodians
Accessibility	Global, open	Requires KYC and has regional limitations
Transparency	On-chain, auditable by anyone	Opaque, private ledgers

Feature	DeFi	CeFi
Control	User-controlled wallets	Institution-controlled accounts
Risks	Code bugs, user error	Central authority failure, high fees

 Benefits of DeFi

- **Open Access:** No approval required from a central authority.
- **Borderless:** Anyone in the world can participate.
- **Higher Yields:** Potential for higher returns through lending and yield farming compared to traditional savings accounts.
- **Innovative:** Enables programmable money and new financial products.

 Limitations and Risks

- **Smart Contract Bugs:** Code can be hacked if not properly audited.
- **Volatility:** The value of crypto assets can change dramatically.
- **No Recovery:** If you lose your private keys, your funds are lost forever.
- ♂ **Lack of Regulation:** Few consumer protections exist.

 Use Cases of DeFi

Use Case	Description
Lending	Lend your crypto assets to earn interest.
Payments	Make instant, cross-border payments.
Staking	Earn rewards by locking tokens to support a blockchain network.
Yield Farming	Move funds across protocols to maximize returns.
NFTs	Create, buy, and sell unique digital assets.
Governance	Vote on protocol upgrades using governance tokens.

 Interactive Questions for Students

- What makes DeFi fundamentally different from traditional banking?
- Can anyone use a DeFi app? Why or why not?
- What is the primary risk if a DeFi smart contract has a bug?
- List two benefits and two risks of using DeFi.

 Wrap-Up
DeFi is revolutionizing finance by replacing intermediaries with code. It offers unprecedented access and innovation but also introduces new risks like smart contract vulnerabilities and a lack of safety nets.

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2. Core DeFi Components

2.1 Lending & Borrowing Protocols

 What Are DeFi Lending and Borrowing Protocols?
In traditional finance, getting a loan involves a bank, paperwork, and a credit check. In DeFi, smart contracts automatically connect lenders and borrowers without any intermediary.

 How Does DeFi Lending Work?

1. **Lenders deposit** crypto (e.g., USDC, ETH) into a smart contract, creating a **lending pool**.
2. **Borrowers can borrow** from this pool by depositing **collateral** (another crypto asset).
3. **Interest is automated:** Borrowers pay interest, and lenders earn interest.
4. When the loan is repaid, the borrower's collateral is returned.

 Example - How Lending Works on Aave

- You deposit $1,000 worth of USDC into Aave. You start earning interest.
- Another user deposits $1,500 worth of ETH as collateral and borrows your $1,000 USDC.
- If the borrower repays the loan plus interest, they get their ETH back.
- If they don't, their ETH collateral can be **liquidated** (sold) to repay the loan.

 What is Collateralization?
Since there are no credit checks, borrowers must provide collateral. DeFi lending requires **over-collateralization**—meaning you must lock up more value than you borrow.

- **Example:** To borrow $100, you might need to lock up $150 worth of ETH as collateral.

  Why Over-Collateralization?
  Crypto prices are volatile. Over-collateralization acts as a safety cushion. If the collateral's value drops, the system still has enough funds to repay the lenders.

  What is Liquidation?
  If the value of the collateral falls below a specific threshold (e.g., the loan is now under-collateralized), the smart contract automatically **liquidates** it. This means the collateral is sold on the market to repay the loan, often with an additional penalty fee for the borrower.

  Key DeFi Lending Ratios

Concept	Definition	Example
Collateral Ratio	Value of collateral ÷ Value of borrowed assets	$150 / $100 = 150%
Liquidation Ratio	The minimum ratio before liquidation occurs	120%
Loan-to-Value (LTV)	Borrowed value ÷ Collateral value	$100 / $150 = 66%

  Real-World Examples of Lending Protocols

- **Aave:** Features "flash loans" and variable interest rates.
- **Compound:** Lenders receive "cTokens" that represent their share and accrue interest.
- **MakerDAO:** A protocol that allows users to generate the DAI stablecoin by locking up collateral.

  Student Reflection / Discussion

- Why do DeFi loans require over-collateralization, while bank loans often do not?
- What happens to a borrower if the price of their collateral drops rapidly?
- How does the liquidation process protect lenders?

2.2 Staking

What is Staking?
Staking is the process of locking up your cryptocurrency to help operate and secure a Proof-of-Stake (PoS) blockchain network. In return, you earn rewards. It's the PoS equivalent of mining in Proof-of-Work.

How Does Staking Work?

1. You **stake** (lock) your tokens in a smart contract or a network node.
2. The blockchain selects **validators** from the stakers to propose and verify new blocks.
3. If you are selected and act honestly, you earn **rewards** (new tokens or transaction fees).
4. If you act maliciously or go offline, your stake can be **slashed** (a penalty where you lose some funds).

Example - Staking ETH on Ethereum

- You can stake 32 ETH to become a full validator, helping to secure the network and earning rewards.
- If you don't have 32 ETH, you can use **staking pools** like Lido or RocketPool, which combine funds from many users.

Why Do People Stake?

Benefits	Risks
Earn passive income	Slashing risk if your validator node fails
Help secure the network	Funds can be locked for a specific period
Support decentralization	Potential for network bugs

Locking & Unstaking

- Some networks have a lock-up period (e.g., Ethereum has a waiting period to unstake).
- **Liquid Staking:** Protocols like Lido give you a tradable token (e.g., `stETH`) representing your staked ETH, allowing you to "use" your locked funds in other DeFi activities.

Types of Staking

Type	Description
Solo Staking	Run your own validator node.
Pooled Staking	Combine funds with others in a pool.
Exchange Staking	Use a centralized service like Coinbase or Binance.
Liquid Staking	Receive a liquid token representing your staked funds.

□ Student Discussion Questions

- Why is staking considered more energy-efficient than mining?
- What are the trade-offs between solo staking and pooled staking?
- How does slashing help keep the network secure?

□ 2.3 Yield Farming

□ What is Yield Farming?

Yield Farming (or Liquidity Mining) is a DeFi strategy where users provide liquidity to a protocol to earn rewards, typically in the form of interest and additional tokens. It's like a high-yield, higher-risk savings account.

□ How Does It Work?

1. You **deposit** crypto assets into a **liquidity pool**.
2. This pool is used by others for trading, lending, or borrowing.
3. You earn a **share of the fees** generated by the pool.
4. You often also earn **reward tokens** (e.g., UNI, COMP) as an extra incentive.

□ Example: Uniswap Yield Farming

- You deposit an equal value of ETH and USDC into a Uniswap liquidity pool.
- You receive **LP Tokens** (Liquidity Provider tokens) that represent your share of the pool.
- As people trade between ETH and USDC, you earn a portion of the 0.3% trading fee.
- Uniswap might also distribute UNI tokens to you as a bonus reward.

□ Key Terms

Term	Description
Liquidity Pool	A smart contract that holds pairs of tokens for trading.
LP Tokens	A receipt that proves your share of the pool.
APR/APY	Annual Percentage Rate / Yield; your potential return.
Impermanent Loss	A potential loss from the value change of the pooled tokens compared to just holding them.

□ Risks of Yield Farming

□ Risk	□ Explanation
Impermanent Loss	The main risk; occurs when the price of your deposited assets changes compared to when you deposited them.
Smart Contract Risk	The protocol's code could have a bug or be exploited by a hacker.
Rug Pulls	Malicious developers can abandon a project and steal investors' funds.
Volatility	The reward tokens you earn can rapidly decrease in value.

□ Popular Yield Farming Platforms

- **Uniswap:** Earn fees from decentralized trading.
- **Aave/Compound:** Earn interest from lending and borrowing.
- **Yearn Finance:** An aggregator that automatically moves your funds to the highest-yielding strategies.

□ Student Discussion Questions

- What is the key difference between staking and yield farming?
- Why do DeFi protocols offer such high rewards to attract liquidity?

- What is impermanent loss, and how can it occur?

2.4 Liquidity Pools

What is a Liquidity Pool?
A Liquidity Pool is a smart contract that holds reserves of two or more tokens, enabling users to trade between them directly without a traditional buyer and seller. It replaces order books with an Automated Market Maker (AMM) system.

How It Works:

- **Liquidity Providers (LPs)** deposit an equal value of two tokens (e.g., ETH and USDC) into the pool.
- The pool uses a mathematical formula (like `x * y = k`) to set prices automatically.
- **Traders** pay a small fee to swap one token for another, and this fee is distributed to all the LPs.

Risk: Impermanent Loss
This is the risk that the value of your share in the pool becomes worth less than if you had just held the two tokens separately. It happens when the price of one token in the pair changes significantly compared to the other.

- **Example:** You deposit ETH and USDC. If the price of ETH doubles, the pool's algorithm rebalances, meaning you would have less ETH and more USDC than you started with. Your pool share might be worth more in USD than initially, but less than if you had just held the ETH.

Popular Liquidity Pool Protocols

- **Uniswap:** The pioneer of AMMs for any ERC-20 token pair.
- **Curve:** Specialized for stablecoin pairs (e.g., USDC/DAI), minimizing impermanent loss.
- **Balancer:** Allows for pools with more than two tokens and customizable weights.

Summary Table

Term	Description
Liquidity Pool	A reserve of tokens used for decentralized trading.

Term	Description
LP Token	A receipt proving your ownership in the pool.
AMM	Automated Market Maker - the algorithm that sets prices.
Impermanent Loss	Loss from price movement of pooled assets vs. holding.

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3. Tokenization & Synthetic Assets

3.1 What is Tokenization?

Definition: Tokenization is the process of converting rights to a real-world asset (like real estate, art, or commodities) into a digital token on a blockchain.

Why Tokenize Assets?

Feature	Explanation
Fractional Ownership	A $1M building can be split into 1M tokens, allowing small investments.
Global Access	Tokens can be bought and sold by anyone, anywhere, 24/7.
Security & Transparency	Ownership and transaction history are immutably recorded on-chain.
Efficiency	Reduces paperwork, middlemen, and speeds up settlement.

Real-Life Example
A commercial building worth $10 million is tokenized into 10 million tokens. An investor can buy 1,000 tokens for $1,000, owning 0.01% of the building and being entitled to 0.01% of its rental income.

3.2 Tokenizing Real-World Assets

A. Real Estate Tokenization

- **How it works:** A property is legally held by a custodian, and ownership rights are represented by tokens on a blockchain.
- **Examples:** Platforms like RealT allow users to buy tokens representing shares in rental properties in the US.
- **Benefits:** Lowers the barrier to entry for real estate investment and provides liquidity.

B. Art Tokenization

- **How it works:** A valuable painting is appraised and its ownership is fractionalized into tokens (often NFTs). Investors can buy "shares" of the artwork.
- **Examples:** Platforms like Maecenas or Masterworks.
- **Benefits:** Democratizes investment in fine art, which was previously only for the ultra-wealthy.

C. Commodities Tokenization (e.g., Gold)

- **How it works:** Physical gold is stored in a secure vault, and each token represents ownership of a specific amount of that gold (e.g., 1 token = 1 gram).
- **Examples:** PAX Gold (PAXG) and Tether Gold (XAUT).
- **Benefits:** Allows for easy, global trading of gold without the hassle of physical storage.

Summary Table

Asset Type	Example	Benefits
Real Estate	RealT	Fractional investing, global liquidity
Artwork	Masterworks	Democratizes fine art investing
Commodities	PAX Gold (PAXG)	Easy access, redeemable for physical gold

Student Reflection Questions

- Why might tokenized real estate be more appealing than traditional property investment?
- What legal challenges could arise when tokenizing a physical asset like art?

3.3 Synthetic Assets

What are Synthetic Assets?
Definition: A synthetic asset (or "synth") is a tokenized derivative that mimics the value of another real-world asset (like Tesla stock or gold) without requiring ownership of the underlying asset.

How Do They Work?
Users lock up **collateral** (usually crypto) in a smart contract to **mint** synthetic assets. The value of the synth is pegged to the target asset via data feeds called **oracles**.

- **Example Workflow:**

1. You lock $150 worth of ETH as collateral.
2. You mint $100 worth of `sTSLA`, a synthetic token that tracks Tesla's stock price.
3. You can now trade `sTSLA` or use it in DeFi. Its value goes up and down with the real TSLA stock.

Key Protocols for Synths

- **Synthetix:** A leading protocol on Ethereum that allows the creation of synths for currencies, commodities, and stocks (sUSD, sBTC, sTSLA).
- **Mirror Protocol:** (Originally on Terra) was designed to create synthetic stocks like mGOOGL (Google) and mTSLA.

Use Cases

- **Access Global Assets:** Someone in a country without access to the US stock market can gain exposure to Apple or Google stock.
- **Arbitrage Trading:** Profit from price differences between the synthetic and the real asset.
- **On-Chain Composability:** Use synthetic stocks as collateral in other DeFi protocols.

3.4 Benefits and Risks of Synthetic Assets

Benefits

Feature	Description
Global Accessibility	No geographic or regulatory barriers to investing.
Permissionless	No brokerage account or approval needed.
Fractional Investment	Buy a fraction of a high-priced stock.
24/7 Trading	Trade anytime, unlike traditional markets.

Risks

Risk	Explanation
Oracle Risk	If the data feed is wrong or manipulated, the synth's value breaks its peg.
Collateral Risk	If the value of the locked collateral crashes, it can trigger mass liquidations.
Regulatory Uncertainty	Tokenizing stocks may attract scrutiny from financial regulators.
Complexity	Requires a good understanding of DeFi mechanics to use safely.

Real-World Example: sTSLA

- Tracks the real Tesla stock price.
- Tradeable 24/7 on decentralized exchanges.
- Does not confer voting rights or dividends from the real Tesla stock.

Student Reflection Questions

- What is the fundamental difference between *tokenizing* gold and creating a *synthetic* gold asset?
- Why is over-collateralization crucial for synthetic asset protocols?

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4. Stablecoins

4.1 What are Stablecoins?

Definition: A stablecoin is a cryptocurrency designed to have a stable value, typically pegged to a fiat currency like the US Dollar (e.g., 1 coin = $1).

Goal: To provide the stability of traditional money with the technological benefits of cryptocurrency (fast, global, programmable).

Analogy: If Bitcoin is a rollercoaster, a stablecoin is a calm ferry ride.

4.2 Types of Stablecoins

Type	Backed By	Example	How It Works
Fiat-backed	Real USD in a bank account	USDT, USDC	The company holds $1 for every 1 token issued.
Crypto-backed	Other cryptocurrencies	DAI	Users lock up more value in crypto (e.g., ETH) to generate stablecoins.
Algorithmic	Algorithms & smart contracts	(UST - defunct)	Code automatically adjusts supply to maintain the peg.

1. Fiat-Backed Stablecoins

- **Mechanism:** Centralized issuers like Tether hold real US dollars in reserve and mint an equivalent number of tokens.
- **Risk:** Requires trust in the issuer to hold the reserves and be transparent. Subject to regulation.

2. Crypto-Backed Stablecoins

- **Mechanism:** Decentralized and over-collateralized. To get $100 of DAI, you might lock up $150 worth of ETH. If ETH's value falls, your position can be liquidated.
- **Risk:** Complex and exposed to the volatility of the underlying crypto collateral.

3. Algorithmic Stablecoins

- **Mechanism:** Uses algorithms to control the token supply. If the price is above $1, new tokens are minted to increase supply and push the price down. If below $1, tokens are burned to reduce supply and push the price up.
- **Risk: Extremely high risk.** This model is fragile and relies on market confidence, which can collapse quickly (see Terra UST case study).

4.3 Use Cases of Stablecoins in DeFi

Use Case	Description
Trading Pair	The primary base currency on DEXs (e.g., ETH/USDC).
Lending & Borrowing	The most common asset to lend or borrow on platforms like Aave.
Payments	Used for fast, cheap, cross-border payments without volatility.
Yield Farming	Often provided as liquidity in farms to earn rewards with lower volatility.
Hedging	A safe haven to park value during market downturns.

 Example: On Compound, you can deposit USDC to earn interest, or borrow USDC by using your ETH as collateral.

4.4 Risks and Famous Collapses

Risks of Stablecoins

Type of Risk	Description
Centralization	Fiat-backed coins require trust in a single company.
Depeg Risk	The stablecoin's value can fall below (or rise above) $1.
Regulatory Pressure	Governments could ban or restrict their use.
Code Vulnerabilities	Crypto-backed and algorithmic stablecoins rely on bug-free smart contracts.